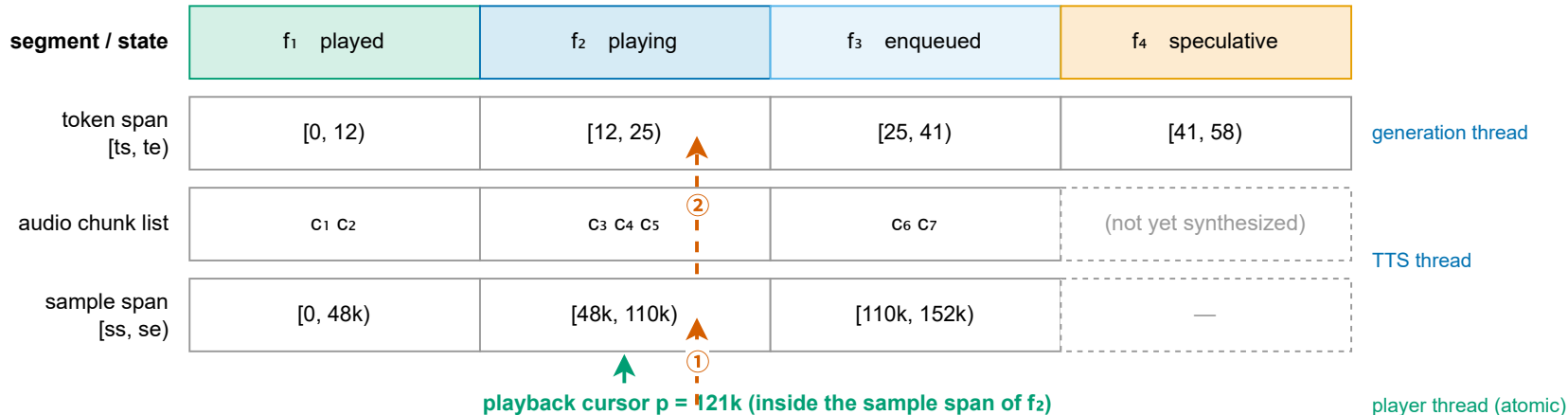


segment records (append-only main axis)



Reverse lookup Φ : ① locate segment f₂ by sample span → ② take te(f₂)=25, yielding H(p). Pure index operation, independent of context length (measured sub-ms). Segments per turn are few: one table lock plus a lock-free playback cursor meets the critical-path requirement.